

Banach Contraction Mapping Theorem

Fixed Point Theory and Applications

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An Introduction to Nonlinear Analysis and Fixed Point Theory

Lecture Notes on Nonlinear Analysis

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Chapter 2b: Banach Contraction Mapping

Complete Metric Spaces and Fixed Point Theorems

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What is a Fixed Point?

Definition (Fixed Point)

Let $T : X \rightarrow X$ be a mapping of a set X into itself. An element $x^* \in X$ is called a **fixed point** (or invariant point) of T if

$$T(x^*) = x^*$$

Example

- $T(x) = x^2$ on $[0, 1]$: fixed points are $x^* = 0$ and $x^* = 1$
- $T(x) = \cos(x)$ on $\mathbb{R}$: fixed point exists (approx. $x^* \approx 0.739$)
- $T(x) = 2x$ on $\mathbb{R}$: no fixed points (except if we allow infinity)

Central Question: When does a mapping have a fixed point?

Motivation: Why Study Fixed Points?

Fixed point theorems have profound applications in mathematics and beyond:

- **Differential Equations:** Existence and uniqueness of solutions
- **Integral Equations:** Solutions via fixed point iteration
- **Numerical Analysis:** Newton-Raphson, iterative methods
- **Economics:** Market equilibrium, game theory
- **Control Theory:** Optimal control problems
- **Fractional Calculus:** Solutions of fractional ODEs

Metric Spaces: Brief Review

Definition (Metric Space)

A **metric space** is a pair (X, d) where X is a set and $d : X \times X \rightarrow [0, \infty)$ is a function (called a metric) satisfying:

- 1 (Non-negativity) $d(x, y) \geq 0$ for all $x, y \in X$
- 2 (Identity) $d(x, y) = 0 \iff x = y$
- 3 (Symmetry) $d(x, y) = d(y, x)$ for all $x, y \in X$
- 4 (Triangle Inequality) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$

Example

- Euclidean space: $(\mathbb{R}^n, d)$ where $d(x, y) = \|x - y\|_2 = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
- Function spaces: $C([a, b])$ with $d(f, g) = \max_{t \in [a, b]} |f(t) - g(t)|$
- Sequence spaces: $\ell^2 = \{(x_n) : \sum_{n=1}^{\infty} x_n^2 < \infty\}$

Convergence and Completeness

Definition (Cauchy Sequence)

A sequence $\{x_n\}$ in a metric space (X, d) is **Cauchy** if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_m, x_n) < \varepsilon \quad \text{for all } m, n > N$$

Definition (Complete Metric Space)

A metric space (X, d) is **complete** if every Cauchy sequence converges to a point in X .

Example (Complete Metric Spaces)

- $(\mathbb{R}^n, \|\cdot\|)$ with any norm
- $C([a, b])$ with uniform norm $\|f\|_\infty = \max |f(t)|$
- Closed bounded intervals $[a, b] \subset \mathbb{R}$
- Any finite dimensional normed space

Definition of Contraction Mapping

Definition (Lipschitz Continuous Mapping)

A mapping $T : X \rightarrow X$ on a metric space (X, d) is **Lipschitz continuous** with constant $L \geq 0$ if

$$d(T(x), T(y)) \leq L \cdot d(x, y) \quad \text{for all } x, y \in X$$

Definition (Contraction Mapping)

A mapping $T : X \rightarrow X$ is called a **contraction** (or **contraction mapping**) if it is Lipschitz continuous with constant $\alpha \in (0, 1)$, i.e.,

$$d(T(x), T(y)) \leq \alpha \cdot d(x, y) \quad \text{for all } x, y \in X, \quad 0 < \alpha < 1$$

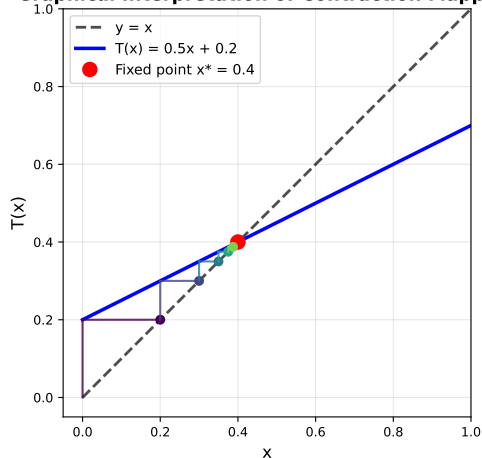
Key Idea: The mapping **shrinks distances** by a factor of $\alpha < 1$.

Example

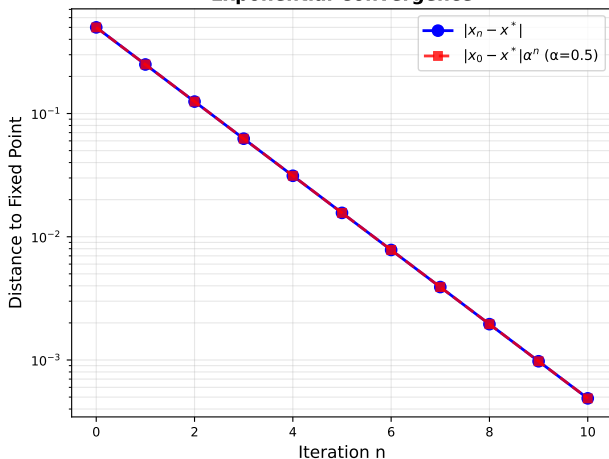
- $T(x) = 0.5x + 0.2$ on $\mathbb{R}$: contraction with $\alpha = 0.5$

Geometric Interpretation

Graphical Interpretation of Contraction Mapping



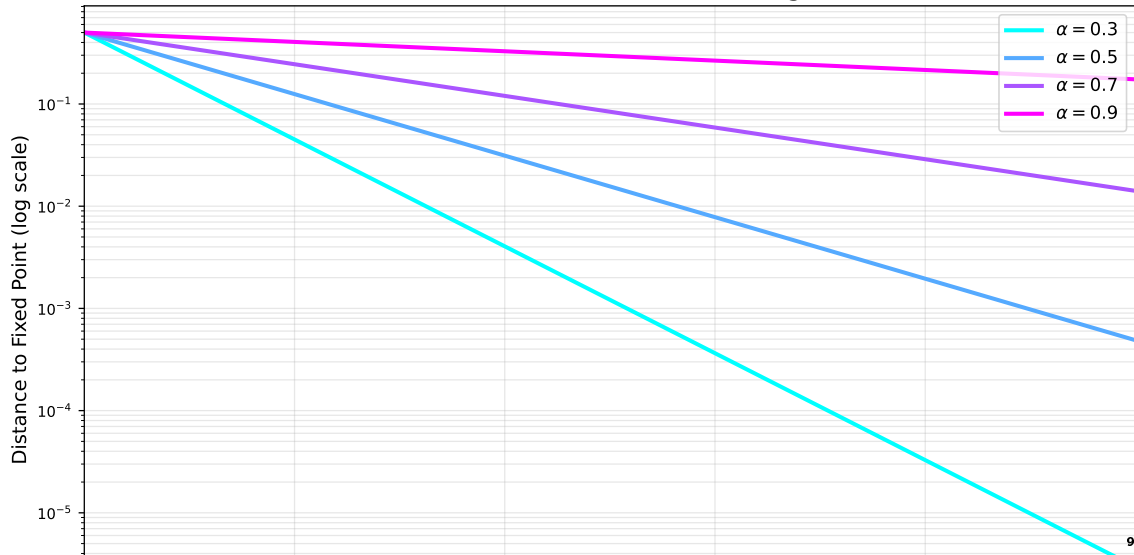
Exponential Convergence



- **Left:** A contraction mapping can be visualized where the function curve is "steeper" than the identity line in some sense

Effect of Contraction Constant

Effect of Contraction Constant on Convergence Rate



Theorem Statement

Theorem (Banach Contraction Mapping Principle)

Let (X, d) be a **complete metric space** and let $T : X \rightarrow X$ be a **contraction mapping** with Lipschitz constant $\alpha \in (0, 1)$. Then:

- 1 T has a **unique fixed point** $x^* \in X$
- 2 For any starting point $x_0 \in X$, the sequence $\{x_n\}$ defined by

$$x_{n+1} = T(x_n), \quad n = 0, 1, 2, \dots$$

converges to x^*

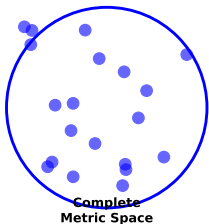
- 3 The rate of convergence is exponential:

$$d(x_n, x^*) \leq \alpha^n d(x_0, x^*)$$

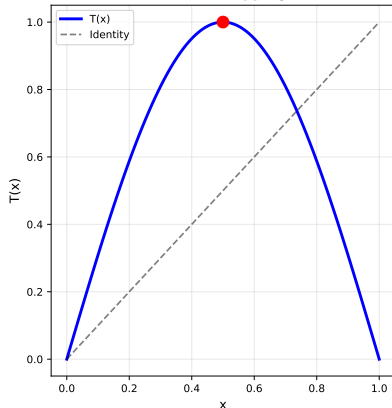
This is one of the most important and useful theorems in analysis!

Three Essential Conditions

(1) Complete Metric Space (X, d)



(2) Continuous Mapping $T: X \rightarrow X$



(3) Contraction Property



$$d(T(x), T(y)) \leq \alpha d(x, y) \\ \text{with } 0 < \alpha < 1$$

BANACH CONTRACTION MAPPING THEOREM (Banach Fixed Point Theorem)

Let (X, d) be a complete metric space and $T: X \rightarrow X$ be a contraction mapping

Sketch of Proof.

❶ **Construction of sequence:** Starting from any $x_0 \in X$, define $x_{n+1} = T(x_n)$.

❷ **Cauchy property:** For $m > n$:

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{m-1}) + \cdots + d(x_{n+1}, x_n) \\ &\leq \alpha^n (1 + \alpha + \cdots + \alpha^{m-n-1}) d(x_1, x_0) \\ &\leq \frac{\alpha^n}{1 - \alpha} d(x_1, x_0) \rightarrow 0 \end{aligned}$$

❸ **Convergence:** Since (X, d) is complete and $\{x_n\}$ is Cauchy, $x_n \rightarrow x^*$ for some $x^* \in X$.

❹ **Fixed point:** By continuity of T :

$$T(x^*) = T(\lim_{n \rightarrow \infty} x_n) = \lim_{n \rightarrow \infty} T(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*$$

❺ **Uniqueness:** If y^* were another fixed point, then

$$d(x^*, y^*) = d(T(x^*), T(y^*)) \leq \alpha d(x^*, y^*)$$

Error Estimation Formulas

The Banach theorem provides explicit error bounds!

Theorem (Error Estimates)

For the Picard iteration $x_{n+1} = T(x_n)$ with contraction $\alpha \in (0, 1)$:

❶ **A posteriori error bound:**

$$d(x_n, x^*) \leq \frac{\alpha}{1 - \alpha} d(x_n, x_{n-1})$$

❷ **A priori error bound:**

$$d(x_n, x^*) \leq \alpha^n d(x_0, x^*)$$

❸ **Error reduction:**

$$d(x_n, x^*) \leq \frac{\alpha^n}{1 - \alpha} d(x_1, x_0)$$

These formulas allow us to:

- Estimate error without knowing the exact fixed point x^*
- Predict how many iterations needed for desired accuracy
- Design stopping criteria

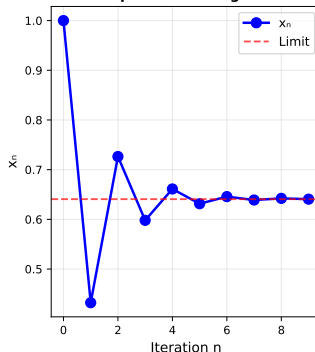
Example: Convergence Analysis

Iteration Table

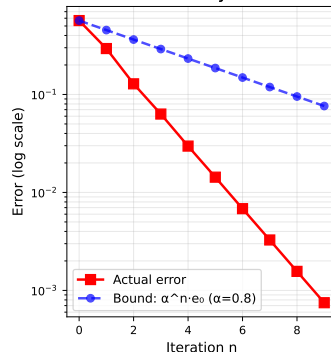
n	x_n	x_{n+1}	$ x_{n+1} - x_n $	Rate
0	1.000000	0.432242	5.68e-01	-
1	0.726423	$T(x_n)$	2.94e-01	0.5181
2	0.598044	$T(x_n)$	1.28e-01	0.4364
3	0.661151	$T(x_n)$	6.31e-02	0.4916
4	0.631429	$T(x_n)$	2.97e-02	0.4710
5	0.645748	$T(x_n)$	1.43e-02	0.4818

Numerical Example: Solving $x = 0.8 \cdot \cos(x)$

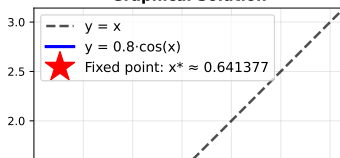
Sequence Convergence



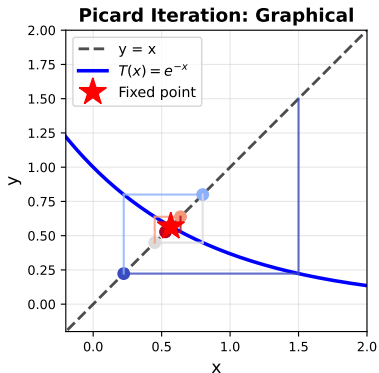
Error Decay Rate



Graphical Solution



Picard Iteration Method



PICARD ITERATION ALGORITHM

Input: Complete metric space (X, d) ,
contraction mapping $T: X \rightarrow X$ with constant $\alpha < 1$,
starting point $x_0 \in X$, tolerance $\varepsilon > 0$

Output: Approximate fixed point x^*

Algorithm:

1. Set $n = 0$
2. While $|x_{n+1} - x_n| > \varepsilon$ do:
 - a) Compute $x_{n+1} = T(x_n)$
 - b) If $|x_{n+1} - x_n| \leq \varepsilon \cdot (1-\alpha)/\alpha$ then
return x_{n+1} (converged)
 - c) Set $n \leftarrow n + 1$
3. If n exceeds max iterations:
warn "Convergence may not be achieved"

Properties:

- Guaranteed convergence
- Exponential error decay
- Error estimate: $|x_n - x^*| \leq \alpha^n |x_0 - x^*|$

Implementation in Python

```
def picard_iteration(T, x0, tol=1e-6, max_iter=100):
    # Picard iteration to find fixed point of T(x)
    x = x0
    errors = []

    for n in range(max_iter):
        x_new = T(x)
        error = abs(x_new - x)
        errors.append(error)

        if error < tol:
            return x_new, n+1, errors

    x = x_new

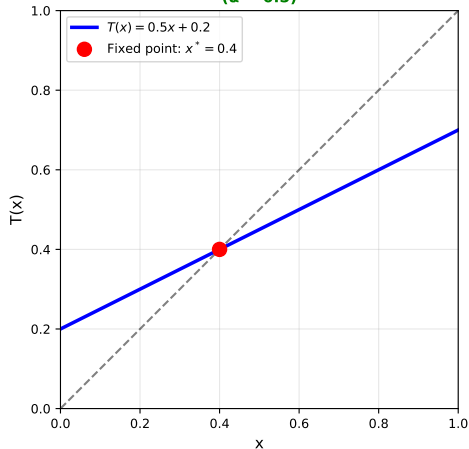
    print("Warning: Not converged after iterations")
    return x, max_iter, errors

# Example: Solve x = cos(x)
T = lambda x: np.cos(x)
x_star, n_iter, errors = picard_iteration(T, x0=0.5)

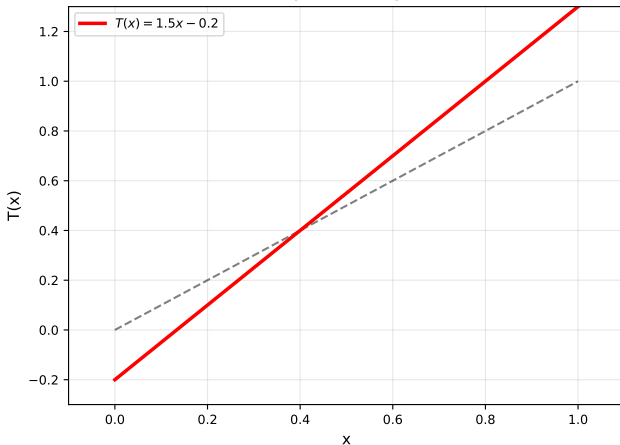
print("Fixed point:", x_star)
print("Iterations:", n_iter)
print("Final error:", errors[-1])
```


Convergence Behavior in Different Cases

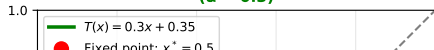
✓ Banach Contraction
($\alpha = 0.5$)



✗ Not a Contraction
($\alpha = 1.5 > 1$)



✓ Stronger Contraction
($\alpha = 0.3$)



Error Decay for Different α Values



Application 1: Existence and Uniqueness for ODEs

Theorem (Picard-Lindelöf Theorem)

Consider the initial value problem (IVP):

$$y' = f(t, y), \quad y(t_0) = y_0$$

If $f(t, y)$ is continuous in t and **Lipschitz continuous in y** with constant L (i.e., $|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$), then on some interval $|t - t_0| \leq h$, there exists a **unique** solution $y(t)$.

Idea.

The IVP is equivalent to the integral equation:

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) \, ds$$

Define $T : C([t_0, t_0 + h]) \rightarrow C([t_0, t_0 + h])$ by $(Ty)(t) = y_0 + \int_{t_0}^t f(s, y(s)) \, ds$.

The Lipschitz condition ensures T is a contraction on an appropriate function space, so Banach's theorem guarantees existence and uniqueness!

Application 2: Solving Nonlinear Equations

Example (Equation: $x = e^{-x}$)

We want to solve $x = e^{-x}$, equivalently find the fixed point of $T(x) = e^{-x}$.

- $T(x) = e^{-x}$ maps $\mathbb{R}$ to $(0, \infty)$
- On $[0, \infty)$: $|T'(x)| = e^{-x} \leq 1$
- On $[0, 1]$: $\sup_{x \in [0, 1]} |T'(x)| = 1 \cdot e^{-0} = 1$
- But we need strict contraction for Banach theorem!
- However, on $[0, 0.5]$: $|T'(x)| = e^{-x} \leq e^{-0} \approx 0.606 < 1$ (OK: contraction!)

Using Picard iteration with $x_0 = 0.5$:

$$x_1 = e^{-0.5} \approx 0.606, \quad x_2 \approx 0.544, \quad x_3 \approx 0.580, \dots \rightarrow x^* \approx 0.567$$

Application 3: Integral Equations

Example (Hammerstein Equation)

Consider the integral equation:

$$x(t) = f(t) + \lambda \int_a^b K(t, s) \phi(s, x(s)) ds$$

Define operator T on $C([a, b])$ by:

$$(Tx)(t) = f(t) + \lambda \int_a^b K(t, s) \phi(s, x(s)) ds$$

If λ is sufficiently small and ϕ is Lipschitz in the second argument, then T is a contraction, and Banach's theorem guarantees unique solution.

- Volterra equations (convolution type)
- Fredholm equations (with small parameter)
- Urysohn equations

Application 4: Numerical Methods

- **Newton-Raphson Method:** Can be viewed as fixed point iteration for $T(x) = x - \frac{f(x)}{f'(x)}$ (when it converges)
- **Successive Approximations:** For systems of equations
- **Iterative Solvers:** For linear systems $Ax = b$, rewrite as $x = x - \alpha(Ax - b)$ and choose α so that the resulting mapping is a contraction
- **Jacobi and Gauss-Seidel Methods:** Iterative solvers for linear systems

The key advantage: **Explicit error bounds** and **guaranteed convergence** with quantified rate!

Generalizations of Banach's Theorem

The classical Banach theorem has been extended in various directions:

- **Quasi-contractions:** $d(Tx, Ty) \leq \alpha d(x, y) + Ld(x, Tx) + Ld(y, Ty)$
- **ϕ -contractions:** $d(Tx, Ty) \leq \phi(d(x, y))$ where ϕ is increasing with $\phi(t) < t$ for $t > 0$
- **Weakly contractive:** $d(Tx, Ty) < d(x, y)$ for all $x \neq y$
- **Asymptotically regular:** $\|T^n(x) - T^{n+1}(x)\| \rightarrow 0$
- **Set-valued contractions:** Extensions to multifunctions
- **Coupled fixed points:** For systems of equations

See Pathak's book Ch. 5 for detailed treatments of:

- Nonexpansive mappings
- Pseudocontractive mappings
- Asymptotically nonexpansive mappings

Comparison with Other Fixed Point Theorems

COMPARISON OF FIXED POINT THEOREMS

THEOREM	SPACE TYPE	CONDITION	CONSTRUCTIVE	RATE	UNIQUENESS
Banach Contraction	Complete Metric	Contraction: $d(Tx, Ty) \leq \alpha d(x, y)$ $\alpha < 1$	YES (Picard) Algorithm available	Linear (α^n)	YES
Brouwer's FPT	Compact Convex Set in $\mathbb{R}^n$	Continuous mapping of X into itself	NO (exists but no algorithm)	No info	Not always
Schauder's FPT	Banach Space	Continuous + Compact image	NO construction	No info	Not always
Kakutani's FPT (Set-valued)	Compact Convex Set	Upper semi- continuous multifunctions	NO (for multi- functions)	No info	Not always

KEY DIFFERENCES:

- **BANACH:** Requires CONTRACTION property → guarantees UNIQUE fixed point
Provides EXPLICIT ALGORITHM with ERROR BOUNDS
Works in any COMPLETE METRIC SPACE (including infinite dimensions)
- **BROUWER:** Only requires CONTINUITY → may have multiple fixed points
Existence only, NO ALGORITHM
Limited to COMPACT CONVEX subsets of $\mathbb{R}^n$
- **SCHAUDER:** Generalizes Brouwer to infinite dimensions
Existence only, NO ALGORITHM
Requires COMPACTNESS condition on the mapping

CHOICE OF THEOREM:

- ✓ Use **BANACH** when: Contraction property is satisfied, uniqueness matters, efficient algorithm needed
- ✓ Use **BROUWER** when: Finite dimensions, continuity sufficient, domain is compact convex subset

When Does Banach Fail?

The Banach theorem requires the contraction condition $d(Tx, Ty) \leq \alpha d(x, y)$ with $\alpha < 1$.

Example (Continuous but not Contractive)

Let $T : [0, 1] \rightarrow [0, 1]$ be defined by $T(x) = x/2$. Then:

- T is continuous and has fixed point $x^* = 0$
- T is a contraction with $\alpha = 1/2$
- So Banach applies! (OK)

Example (Continuous but NOT Contractive)

Let $T : [0, 1] \rightarrow [0, 1]$ be the Brouwer example: $T(x) = \sqrt{x}$. Then:

- T is continuous with fixed points $x^* = 0$ and $x^* = 1$
- $|T'(x)| = \frac{1}{2\sqrt{x}}$ unbounded as $x \rightarrow 0^+$
- So T is NOT a uniform contraction
- But Brouwer's fixed point theorem still guarantees existence!

Example 1: Solving $x = \cos(x)$

```
import numpy as np

# Define the function and its derivative
T = lambda x: np.cos(x)
T_prime = lambda x: -np.sin(x)

# Check contraction on [0, pi/2]
alpha = np.max(np.abs(T_prime(np.linspace(0, np.pi/2, 100))))
print("Contraction constant alpha ~= ", alpha)

# Picard iteration
x = 0.5
for n in range(10):
    x_new = T(x)
    error = abs(x_new - x)
    print("x_%d = %f, error = %e" % (n, x, error))
    x = x_new

print("Fixed point:", x)
print("Verification: cos(x) =", np.cos(x))
```

Example 2: System via Fixed Point

```
import numpy as np

def T(z):
    x, y = z
    x_new = 0.5 * np.sin(x) + 0.5 * np.cos(y)
    y_new = 0.5 * np.cos(x) + 0.3 * x
    return np.array([x_new, y_new])

# Initial guess
z = np.array([0.5, 0.5])

# Iteration
for n in range(50):
    z_new = T(z)
    error = np.linalg.norm(z_new - z)
    if error < 1e-8:
        print("Converged at iteration", n)
        break
    z = z_new

print("Solution: x =", z[0], "y =", z[1])
print("Verification: T(z) =", T(z))
```

Exercise 1

Definition (Exercise)

Show that $T(x) = \frac{1}{2} \sin(x) + \frac{1}{4}$ is a contraction on $\mathbb{R}$. Find the contraction constant α and estimate the number of iterations needed to achieve accuracy $\varepsilon = 10^{-6}$ starting from $x_0 = 0$.

Solution hint:

- 1 Find $\alpha = \sup_{x \in \mathbb{R}} |T'(x)| = \sup_{x \in \mathbb{R}} \frac{1}{2} |\cos(x)| = \frac{1}{2}$
- 2 Use error estimate: $d(x_n, x^*) \leq \alpha^n d(x_0, x^*)$
- 3 Need $\alpha^n < \varepsilon$, so $n > \frac{\log \varepsilon}{\log \alpha}$
- 4 Compute: $n > \frac{\log 10^{-6}}{\log 0.5} \approx 19.93$, so $n \geq 20$ iterations

Exercise 2

Definition (Exercise)

For the IVP $y' = y + t$, $y(0) = 1$, use Picard iteration to construct the first few approximations:

$$y_n(t) = 1 + \int_0^t (y_{n-1}(s) + s) ds$$

starting with $y_0(t) = 1$.

Solution:

$$y_0(t) = 1$$

$$y_1(t) = 1 + \int_0^t (1 + s) ds = 1 + t + \frac{t^2}{2}$$

$$y_2(t) = 1 + \int_0^t \left(1 + s + \frac{s^2}{2} + s\right) ds = 1 + 2t + \frac{3t^2}{4} + \frac{t^3}{6}$$

$\vdots$

Exercise 3

Definition (Exercise)

For $T(x) = 0.4x + 0.3$, find:

- 1 The fixed point x^*
- 2 The contraction constant α
- 3 The number of iterations to reach $|x_n - x^*| < 10^{-10}$ starting from $x_0 = 0$
- 4 The actual sequence $\{x_n\}$ for $n = 0, 1, \dots, 10$

Solution:

- 1 Fixed point: $x^* = 0.4x^* + 0.3 \Rightarrow x^* = \frac{0.3}{0.6} = 0.5$
- 2 Contraction: $\alpha = |T'(x)| = 0.4$
- 3 Iterations: $0.4^n \cdot 0.5 < 10^{-10} \Rightarrow n > \frac{\log(0.5 \times 10^{-10})}{\log 0.4} \approx 24.3$, so $n = 25$
- 4 Sequence: $x_0 = 0, x_1 = 0.3, x_2 = 0.42, x_3 = 0.468, \dots \rightarrow 0.5$

Key Takeaways

- 1 **Fixed Point Problem:** Find x^* such that $T(x^*) = x^*$
- 2 **Banach Theorem:** In a complete metric space, contractions have unique fixed points with exponential convergence
- 3 **Constructive Method:** Picard iteration $x_{n+1} = T(x_n)$ provides explicit algorithm
- 4 **Error Bounds:** Explicit formulas allow us to estimate errors without knowing x^*
- 5 **Applications:** Existence/uniqueness for ODEs, integral equations, numerical methods, etc.
- 6 **Necessity of Conditions:** All three conditions (complete space, contraction, self-mapping) are essential
- 7 **Generalizations:** Extended to weaker conditions, set-valued mappings, infinite dimensions

The Banach Contraction Mapping Theorem is a cornerstone of analysis and numerical mathematics!

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-  Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-181.
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-  Kirk, W. A., & Sims, B. (Eds.). (2001). *Handbook of Metric Fixed Point Theory*. Kluwer Academic Publishers.

Additional Resources

- Pathak, H. K. (2018). Chapter 5: Fixed Point Theorems - detailed treatment of contractions and generalizations
- Online courses: MIT 18.100 (Real Analysis), Stanford CME 100 (Scientific Computing)
- Interactive visualizations: Fixed point iteration applets, Desmos graphing
- Numerical algorithms: Implementation in Python (`scipy.optimize`), MATLAB
- Applications: Numerical methods textbooks (Burden and Faires, Atkinson)